
Variational Inference

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Outline

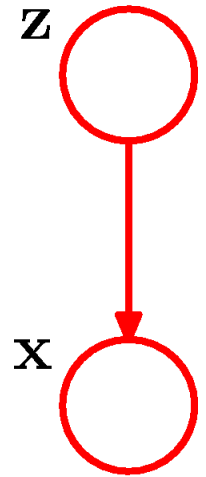
- GMM with Latent variables
- Variational Inference

GMM with Latent Variables

Gaussian Mixture Model

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k N(\mathbf{x} | \mu_k, \Sigma_k)$$

- In GMM, we can adopt K-dimensional binary random variable $\mathbf{z}$
- Simply, we would say this random variable represents the “effect” of k ’th Gaussian component
- ex) $\mathbf{z} = \{1, 0, 0, \dots, 0, 0\}$
- Each element denotes the choice of each Gaussian component
- Let π_k be the probability of choosing each Gaussian component
- ex) $p(\mathbf{z} = \{1, 0, 0, \dots, 0, 0\}) = \pi_1$
- $p(\mathbf{z} = \{0, 1, 0, \dots, 0, 0\}) = \pi_2$



GMM with Latent Variables

Gaussian Mixture Model

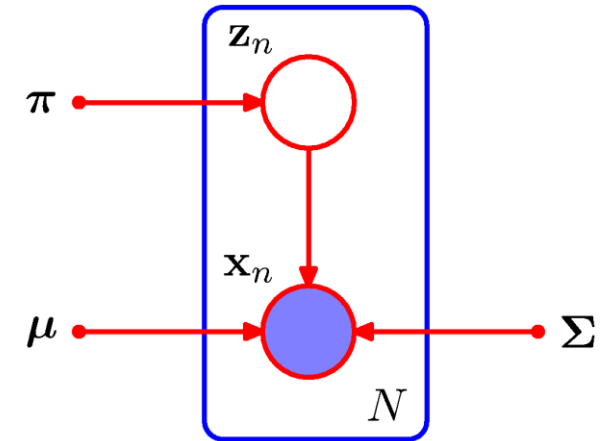
$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k N(\mathbf{x} | \mu_k, \Sigma_k)$$

- ex) $\mathbf{z} = \{1, 0, 0, \dots, 0, 0\}$
- ex) $p(\mathbf{z} = \{1, 0, 0, \dots, 0, 0\}) = \pi_1$
- $p(\mathbf{z} = \{0, 1, 0, \dots, 0, 0\}) = \pi_2$
- Consequently, latent random variable $\mathbf{z}$ has the characteristics as follows:

$$p(z_k = 1) = \pi_k$$

$$0 \leq \pi_k \leq 1$$

$$\sum_{k=1}^K \pi_k = 1$$



GMM with Latent Variables

- Then we can derive the formula as below, for the distribution of the latent variable $\mathbf{z}$:

$$p(z_k = 1) = \pi_k$$

$$p(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k}$$

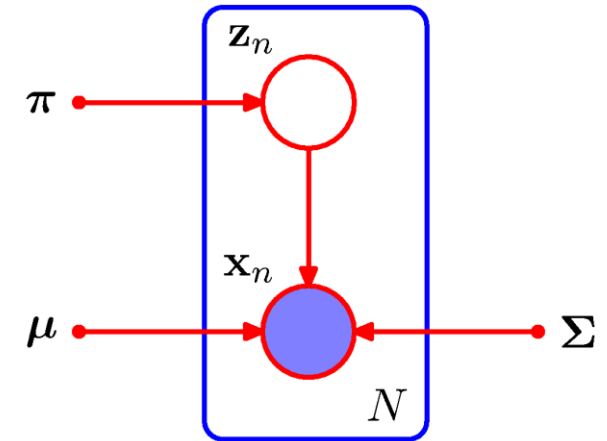
$$p(\mathbf{x}|z_k = 1) = N(\mathbf{x}|\mu_k, \Sigma_k)$$

$$p(\mathbf{x}|\mathbf{z}) = \prod_{k=1}^K N(\mathbf{x}|\mu_k, \Sigma_k)^{z_k}$$

- The marginal distribution of $\mathbf{x}$ is given as:

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z}) = \sum_{k=1}^K \pi_k N(\mathbf{x}|\mu_k, \Sigma_k)$$

- which is the same formulation of GMM in the last lecture



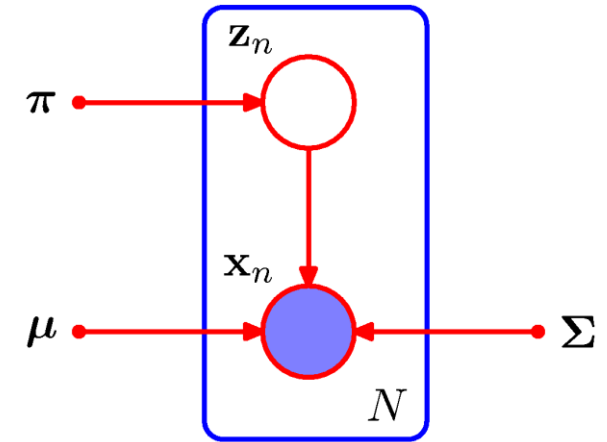
GMM with Latent Variables

$$p(z_k = 1) = \pi_k$$

$$p(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k}$$

$$p(\mathbf{x}|z_k = 1) = N(\mathbf{x}|\mu_k, \Sigma_k)$$

$$p(\mathbf{x}|\mathbf{z}) = \prod_{k=1}^K N(\mathbf{x}|\mu_k, \Sigma_k)^{z_k}$$



Using the distribution of the Latent variable above, we can compute the posteriori (responsibilities) when datapoint $\mathbf{x}$ is given:

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{p(z_k = 1)p(\mathbf{x}|z_k = 1)}{\sum_j^K p(z_j = 1)p(\mathbf{x}|z_j = 1)} = \frac{\pi_k N(\mathbf{x}|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(\mathbf{x}|\mu_j, \sigma_j)}$$

GMM with Latent Variables

EM algorithm of likelihood with Latent variable

Suppose $\mathbf{X}$ and $\mathbf{Z}$ are observed, then the log likelihood is given as:

$$\ln p(\mathbf{X}, \mathbf{Z} | \theta)$$

But in many cases latent variable $\mathbf{Z}$ cannot be observed, we compute the marginal distribution by considering all possibilities of latent variables:

$$\ln p(\mathbf{X} | \theta) = \ln \left\{ \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z} | \theta) \right\}$$

- Note that we have log-sum expression
 - Even we assume that Joint probability belongs to Exponential family, this marginal distribution does not.
 - This cannot be expressed as the simple quadratic function
 - which requires complex MLE solve
-

GMM with Latent Variables

$$\ln p(\mathbf{X}|\theta) = \ln \left\{ \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta) \right\}$$

Since the complex form of marginal distribution is hard to handle for MLE, we try to use Expectation of loglikelihood as the following: (why?)

$$E_Z[\ln p(\mathbf{X}, \mathbf{Z}|\theta)] = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$$

By adopting Expectation-maximization to this Expectation of log-likelihood, we can approximately have the MLE solution

GMM with Latent Variables

EM algorithm of the likelihood with Latent variable

$$\ln p(\mathbf{X}|\theta) = \ln \left\{ \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta) \right\}$$

E-step

- first let every parameters be the constants, say θ^{old}
- compute the probability distribution $p(\mathbf{Z}|\mathbf{X}, \theta^{old})$

$$E_{\mathbf{Z}}[\ln p(\mathbf{X}, \mathbf{Z}|\theta)] = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$$

$$Q(\theta, \theta^{old}) = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$$

GMM with Latent Variables

EM algorithm of the likelihood with Latent variable

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M-step

- using the posteriori of Z below, compute the new parameter which maximize Q

$$\theta^{new} = \arg \max_{\theta} Q(\theta, \theta^{old})$$

GMM with Latent Variables

Latent variable을 도입한 likelihood의 EM algorithm

EM 알고리즘 일반화

- 관찰데이터 $\mathbf{X}$ 와 잠재 변수 $\mathbf{Z}$ 가 파라미터 θ 에 의해 주어졌을 때,
- 결합 분포는 $p(\mathbf{X}, \mathbf{Z}|\theta)$ 와 같이 표현 가능하다.
- 이 때 $p(\mathbf{X}|\theta)$ 값을 가장 크게 만드는 파라미터 θ 값을 얻고 싶다. (MLE를 이용)
 - **Init Step** : (임의의) 파라미터 θ^{old} 의 값을 설정한다.
 - **E-Step** : $p(\mathbf{Z}|\mathbf{X}, \theta^{old})$ 값을 추론한다.
 - **M-Step** : θ^{new} 값을 추론한다. 이 때,
 - $\theta^{new} = \arg \max_{\theta} Q(\theta, \theta^{old})$
 - $Q(\theta, \theta^{old}) = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$
 - 새롭게 구해진 파라미터의 값들이 수렴 상태인지 확인한다.
 - 수렴되지 않았다면 아래 작업 후 Step-2 로 돌아간다.
 - 수렴되었을 경우 종료한다.
 - $\theta^{old} \leftarrow \theta^{new}$

$$\ln p(\mathbf{X}|\theta) = \ln \left\{ \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta) \right\}$$

$$E_Z[\ln p(\mathbf{X}, \mathbf{Z}|\theta)] = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$$

GMM revisited

Gaussian Mixture Model

$$\ln p(\mathbf{X}|\mu, \Sigma, \pi) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$$

Here, latent variable is not directly expressed, but we can adopt latent variable to perform EM algorithm

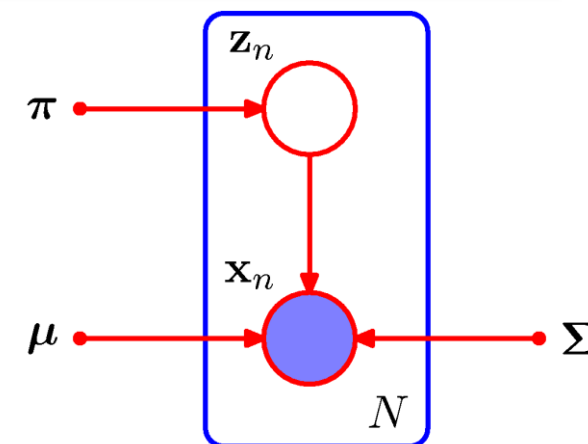
1) E-step: compute $p(\mathbf{Z}|\mathbf{X}, \theta^{old})$

$$p(\mathbf{Z}|\mathbf{X}, \mu, \Sigma, \pi) \propto \prod_{n=1}^N \prod_{k=1}^K [\pi_k N(\mathbf{x}_n | \mu_k, \Sigma_k)]^{z_{nk}}$$

where

$$p(\mathbf{z}) = \prod_{k=1}^K \pi^{z_k}$$

$$p(\mathbf{x}|\mathbf{z}) = \prod_{k=1}^K N(\mathbf{x}|\mu_k, \Sigma_k)^{z_k}$$



GMM revisited

Gaussian Mixture Model

$$\ln p(\mathbf{X}|\mu, \Sigma, \pi) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$$

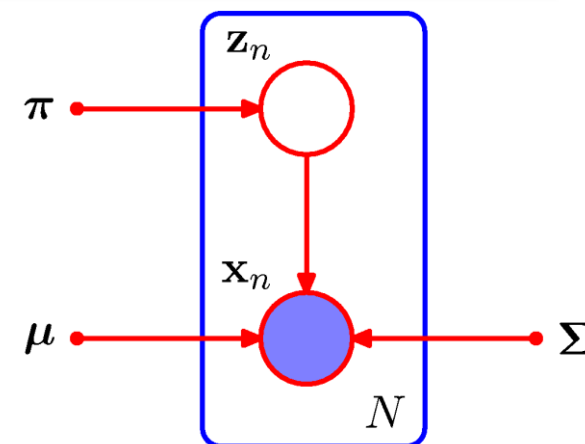
Here, latent variable is not directly expressed, but we can adopt latent variable to perform EM algorithm

2) M-step: compute optimal parameter

$$Q(\theta, \theta^{old}) = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$$

$$\theta^{new} = \arg \max_{\theta} Q(\theta, \theta^{old})$$

$$Q(\theta, \theta^{old}) = E_{\mathbf{Z}} [\ln p(\mathbf{X}, \mathbf{Z}|\mu, \Sigma, \pi)] = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) \{ \ln \pi_k + \ln N(\mathbf{x}_n | \mu_k, \Sigma_k) \}$$



GMM revisited

$$\begin{aligned} E_{\mathbf{Z}}[\ln p(\mathbf{X}, \mathbf{Z}|\theta)] &= E_{\mathbf{Z}}\left[\sum_n \ln p(\mathbf{x}_n, \mathbf{z}_n|\theta)\right] = \sum_n E_{\mathbf{Z}}[\ln p(\mathbf{x}_n, \mathbf{z}_n|\theta)] \\ &= \sum_{n=1}^N E_{\mathbf{Z}}[\ln(p(\mathbf{z}_n)p(\mathbf{x}_n|\mathbf{z}_n, \theta))] = \sum_{n=1}^N E_{\mathbf{Z}}\left[\ln\left[\prod_{k=1}^K (\pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k))^{z_{nk}}\right]\right] \\ &= \sum_{n=1}^N \sum_{k=1}^K E_z[(z_{nk}) \ln(\pi_k N(\mathbf{x}|\mu_k, \Sigma_k))] = \sum_{n=1}^N \sum_{k=1}^K E_z[z_{nk}] \ln(\pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k)) \\ &= \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) \{\ln \pi_k + \ln N(\mathbf{x}_n|\mu_k, \Sigma_k)\} \end{aligned}$$

$$\begin{aligned} p(\mathbf{z}) &= \prod_{k=1}^K \pi^{z_k} \\ p(\mathbf{x}|\mathbf{z}) &= \prod_{k=1}^K N(\mathbf{x}|\mu_k, \Sigma_k)^{z_k} \end{aligned}$$

$$E_Z[\ln p(\mathbf{X}, \mathbf{Z}|\theta)] = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$$

where

$$\begin{aligned} E[z_{nk}] &= \frac{\sum_{\mathbf{z}_n} z_{nk} \prod_{k'} [\pi_{k'} N(\mathbf{x}_n|\mu_{k'}, \Sigma_{k'})]^{z_{nk'}}}{\sum_{\mathbf{z}_n} \prod_j [\pi_j N(\mathbf{x}_n|\mu_j, \Sigma_j)]^{z_{nj}}} \\ &= \frac{\pi_k N(\mathbf{x}_n|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(\mathbf{x}_n|\mu_j, \Sigma_j)} = \gamma(z_{nk}) \end{aligned}$$

GMM revisited

$$Q(\theta, \theta^{old}) = E_{\mathbf{Z}} [\ln p(\mathbf{X}, \mathbf{Z} | \mu, \Sigma, \pi)] = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) \{ \ln \pi_k + \ln N(\mathbf{x}_n | \mu_k, \Sigma_k) \}$$

By applying partial derivation (see last lecture) to perform MLE, we have:

$$\mu_k = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n$$

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k)(\mathbf{x}_n - \mu_k)^T$$

$$\pi_k = \frac{N_k}{N}$$

Outline

- GMM with Latent variables
- Variational Inference

Variational Inference

Suppose we have the data X of the model which involves Latent variable Z . The marginal distribution is given as:

$$p(\mathbf{X}|\theta) = \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta)$$

- expressed as summation when the case of Discrete latent variable
- integral for the case of Continuous latent variable

Now, suddenly, suppose we are going to use $q(\mathbf{Z}|\theta')$, which is the distribution of Z . Then let us consider the following identity expressed with $q(\mathbf{Z}|\theta')$:

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q||p)$$

Variational Inference

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q||p)$$

Where

$$L(q, \theta) = \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\}$$
$$KL(q||p) = - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)}{q(\mathbf{Z})} \right\}$$

(not important)

- the above equations are **functionals**, which take the functions as their domain
(ex: entropy, KL-divergence, Fourier transform)
 - However, L can be represented as $L(\theta; q)$ which is the function of real value of the parameter θ
-

Variational Inference

Prior knowledge

- **Jensen's Inequality** (옌슨 부등식)

- 만약 $f(x)$ 가 **convex** 를 만족한다면 다음이 항상 성립한다.

$$E[f(x)] \geq f(E[x])$$

- **Gibb's Inequality** (깁스 부등식)

- 함수 q 와 p 에 대해 $KL[p||q] \geq 0$ 을 항상 만족한다.

- (참고) 만약 $p = q$ 이면 $KL[p||q] = 0$

- (참고) 만약 $p \neq q$ 이면 $KL[p||q] > 0$

- (증명) $KL[p||q] = \sum_i p_i \ln \frac{p_i}{q_i} = - \sum_i p_i \ln \frac{q_i}{p_i} \geq - \ln[\sum_i p_i \frac{q_i}{p_i}] = - \ln[\sum_i q_i] = 0$

Variational Inference

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q||p)$$

Where

$$L(q, \theta) = \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\}$$

$$KL(q||p) = - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)}{q(\mathbf{Z})} \right\}$$

Variational Inference

$$\ln p(\mathbf{X}|\theta) = \ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta)$$

L can be obtained by adopting $q(\mathbf{z})$:

$$\ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta) = \ln \sum_{\mathbf{Z}} q(\mathbf{Z}) \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \geq \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} = L(q, \theta)$$

**Jensen's
Inequality**

The above $L(\theta; q)$ is the lower bound
(Evidence Lower BOund, ELBO) of the log likelihood $\ln p(\mathbf{X}|\theta)$ (why?)

Variational Inference

Proof) let us compute the difference btwn log likelihood $\ln p(\mathbf{X}|\theta)$ and $L(\theta; q)$:

$$\begin{aligned}\ln p(\mathbf{X}|\theta) - L(q, \theta) &= \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} \\ &= \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)p(\mathbf{X}|\theta)}{q(\mathbf{Z})} \right\} \\ &= \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)}{q(\mathbf{Z})} \right\} - \ln p(\mathbf{X}|\theta) \sum_{\mathbf{Z}} q(\mathbf{Z}) \\ &= - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)}{q(\mathbf{Z})} \right\} = KL[q(\mathbf{Z}) \| p(\mathbf{Z}|\mathbf{X}, \theta)] = KL[q \| p]\end{aligned}$$

finally we have:

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q \| p) \boxed{\geq} L(\theta; q)$$

**Gibb's
Inequality**

Variational Inference

$$\begin{aligned}\ln p(\mathbf{X}|\theta) &= L(q, \theta) + KL(q||p) \\ &= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} + KL[q(\mathbf{Z})||p(\mathbf{Z}|\mathbf{X}, \theta)]\end{aligned}$$

Variational Inference

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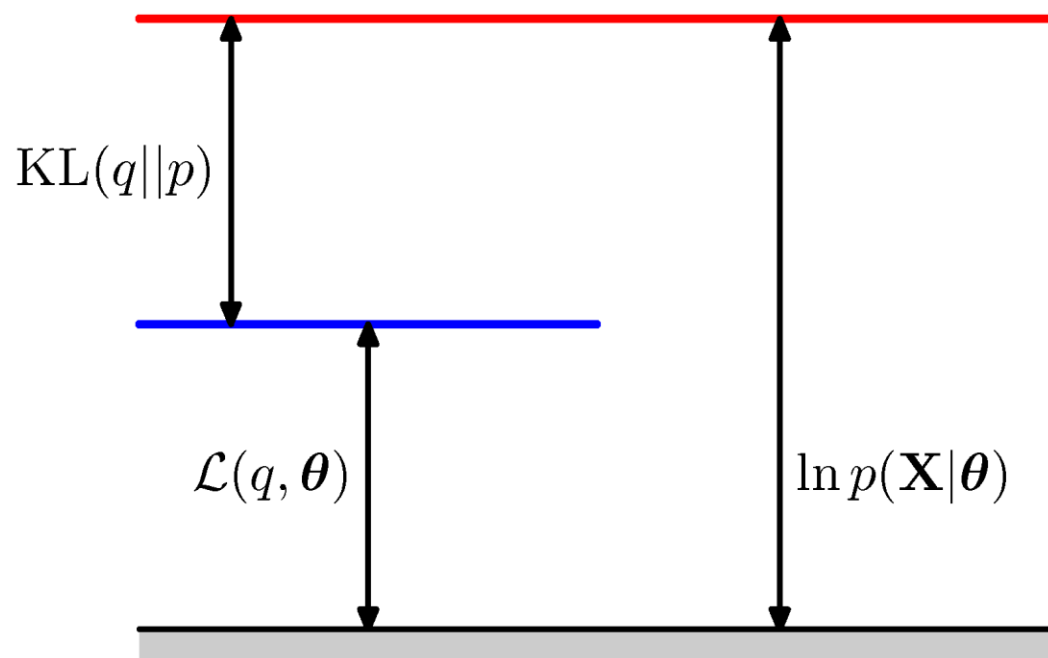
Posterior of
Latent variables

Variational Inference

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q||p)$$

$$= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} + KL[q(\mathbf{Z}) || p(\mathbf{Z}|\mathbf{X}, \theta)]$$

Posterior of
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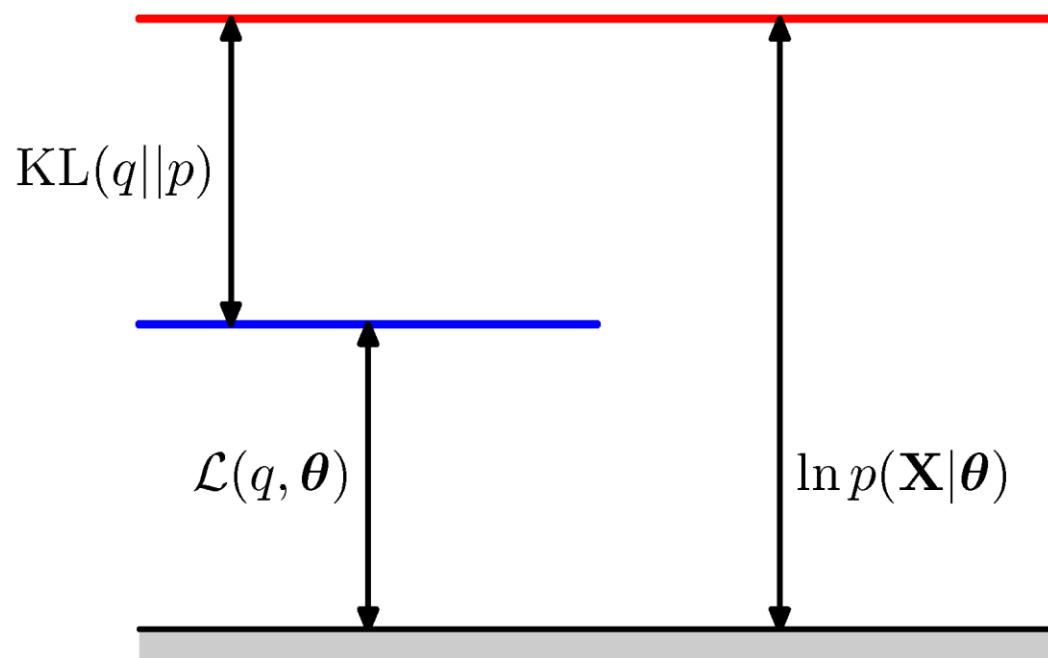
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Posterior of
Latent variables

- The essence of Variational inference lies in finding a function $q(\mathbf{Z}|\theta')$ that can approximate the posterior of latent variables which is difficult to approximate



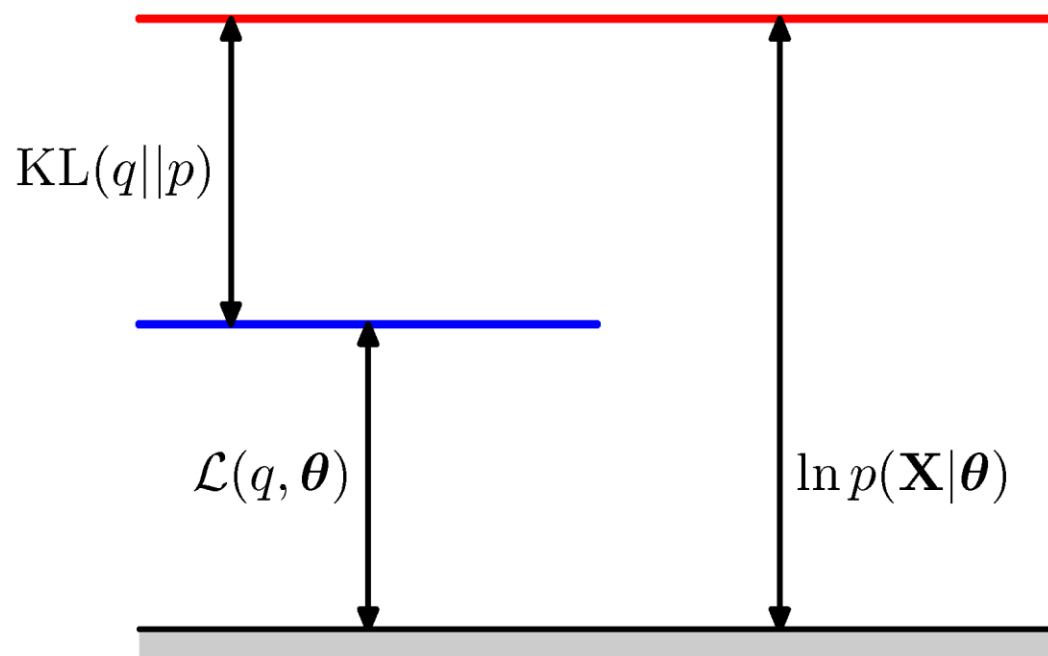
Variational Inference

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Posterior of
Latent variables

- The essence of Variational inference lies in finding a function $q(\mathbf{Z}|\theta')$ that can approximate the posterior of latent variables which is difficult to approximate
- Instead of directly computing the intractable posterior and KL directly, we approximately find $q(\mathbf{Z}|\theta')$ by maximizing the easily computable ELBO
- Of course, apparently if it is easier to directly compute the KL, there is no need to compute ELBO



Variational Inference

$$\begin{aligned}\ln p(\mathbf{X}|\theta) &= L(q, \theta) + KL(q||p) \\ &= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} + KL[q(\mathbf{Z})||p(\mathbf{Z}|\mathbf{X}, \theta)]\end{aligned}$$

- We can express ELBO as the following:

$$\begin{aligned}L(q, \theta) &= ELBO = \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(X, Z|\theta)}{q(Z)} \right\} \\ &= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \frac{p(X|Z, \theta)p(Z)}{q(Z)} \\ &= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln p(X|Z, \theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \frac{q(Z)}{p(Z)} \\ &= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln p(X|Z, \theta) - KL(q(Z)||p(Z))\end{aligned}$$

- what if we do not know not only $q(\mathbf{Z}|\theta')$, but also $p(X|Z, \theta)$ and $p(Z)$ (what if θ is unknown)?
 - => use EM algorithm
-

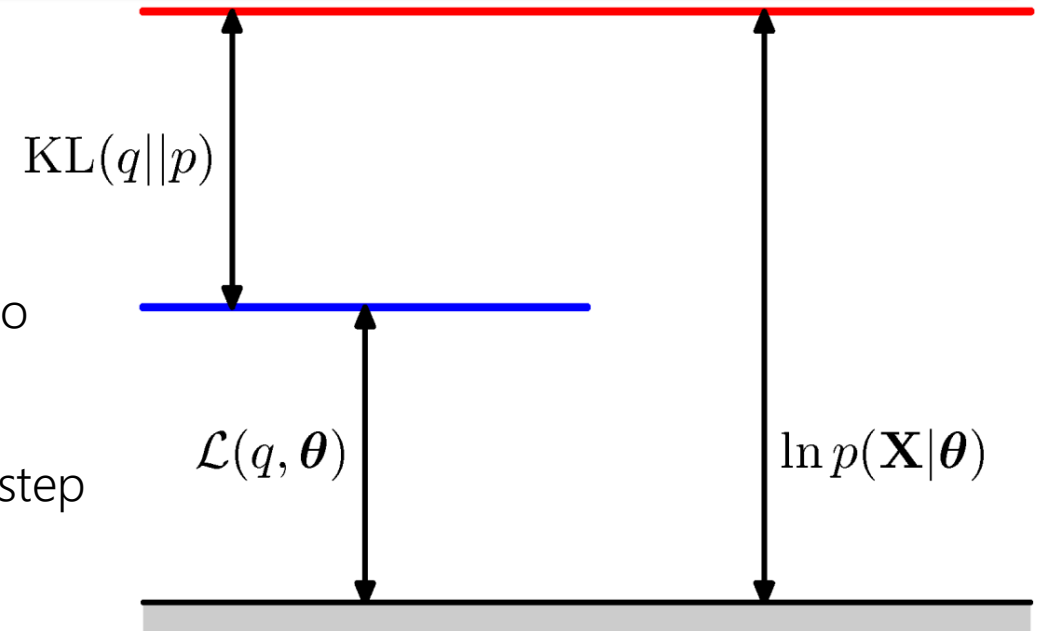
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Variational Inference

E-step

- we are going to minimize KL-divergence term
- In this step we fix θ (θ^{old})
- one can consider θ as being in a fixed state among the two variables of $L(\theta; q)$
- In other words, during the E-step, it can be viewed as the step of estimating q (approximating the posterior $p(z|x)$) that maximizes $L(\theta^{old}; q)$



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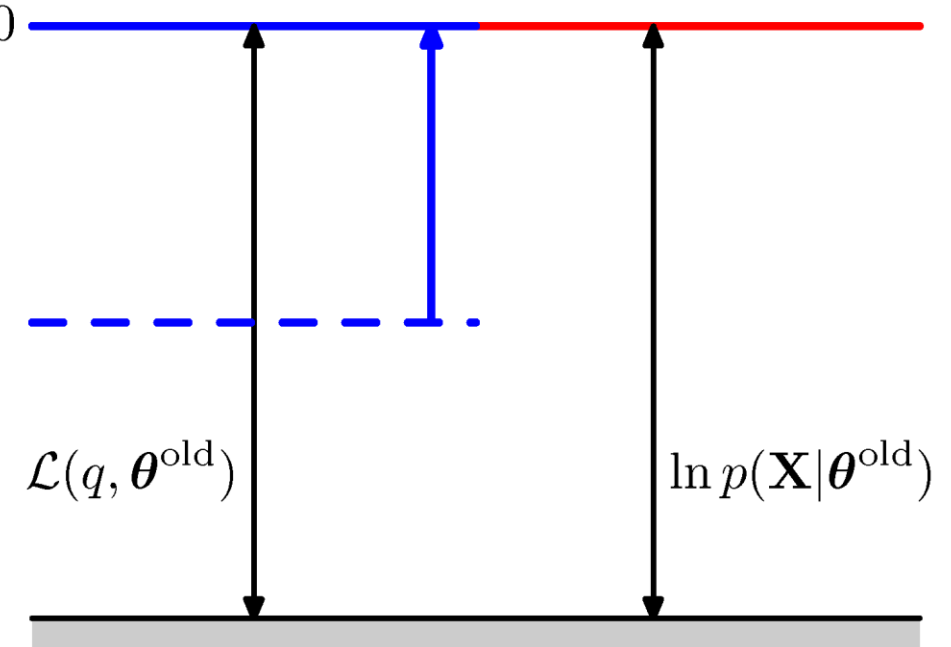
$$= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} + KL[q(\mathbf{Z})||p(\mathbf{Z}|\mathbf{X}, \theta)]$$

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$$KL(q||p) = 0$$



$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q||p)$$

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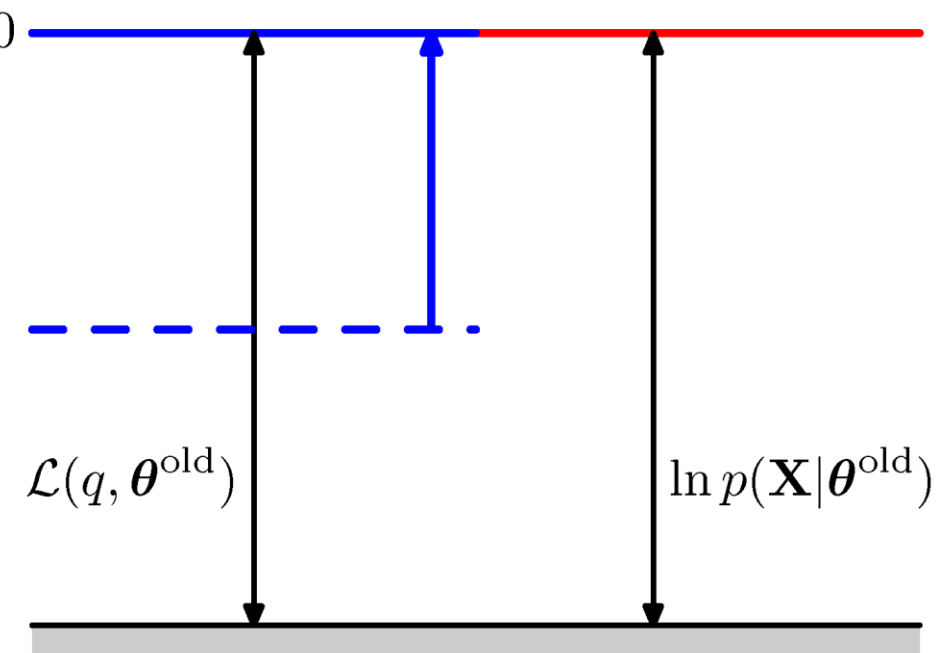
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- In other words, during the E-step, it can be viewed as the step of estimating q (approximating the posterior $p(z|x)$) that maximizes $L(\theta^{old}; q)$

- In the case of GMM, the posterior is tractable, thereby there is no need to redefine $q(\mathbf{Z}|\theta')$ to find θ' . We just simple choose $q = p(z|x, \theta^{old})$ as the posterior.

$$KL(q||p) = 0$$



$$p(\mathbf{Z}|\mathbf{X}, \mu, \Sigma, \pi) \propto \prod_{n=1}^N \prod_{k=1}^K [\pi_k N(\mathbf{x}_n | \mu_k, \Sigma_k)]^{z_{nk}}$$

Variational Inference

M-step

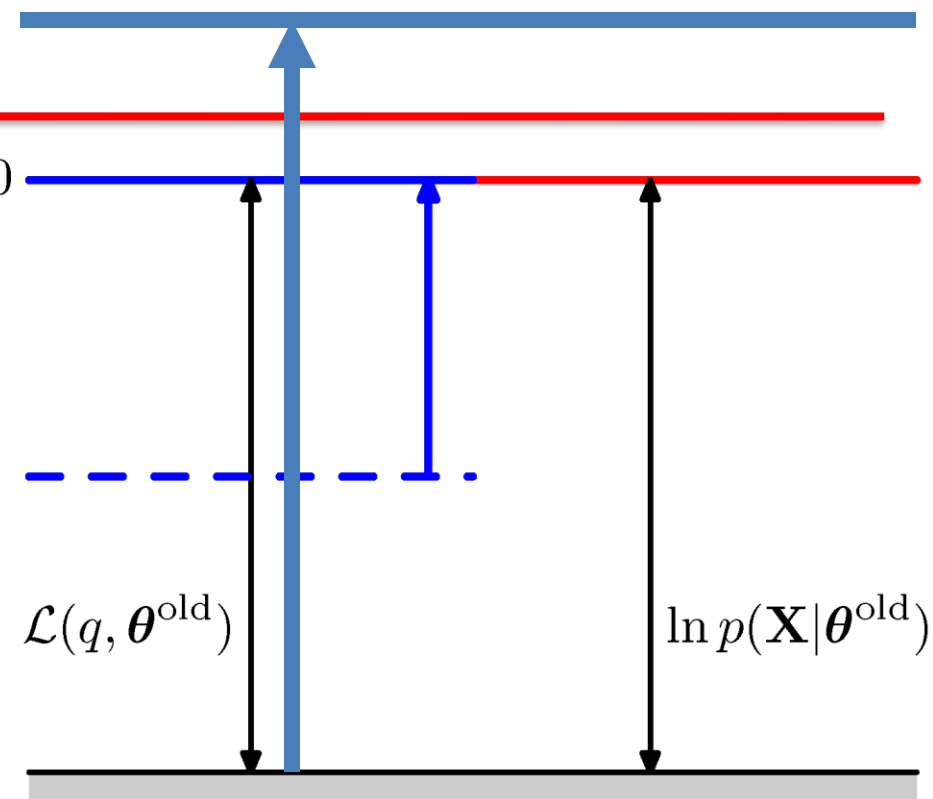
- In this step, by using $p(\mathbf{z}|\mathbf{x}, \theta^{old})$ we have the following solution:

$$Q(\theta, \theta^{old}) = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$$

$$\theta^{new} = \arg \max_{\theta} Q(\theta, \theta^{old})$$

Fixing q is analogous to estimating a new parameter θ^{new} that maximizes the ELBO

$$\text{KL}(q||p) = 0$$



Variational Inference

M-step

- In summary:

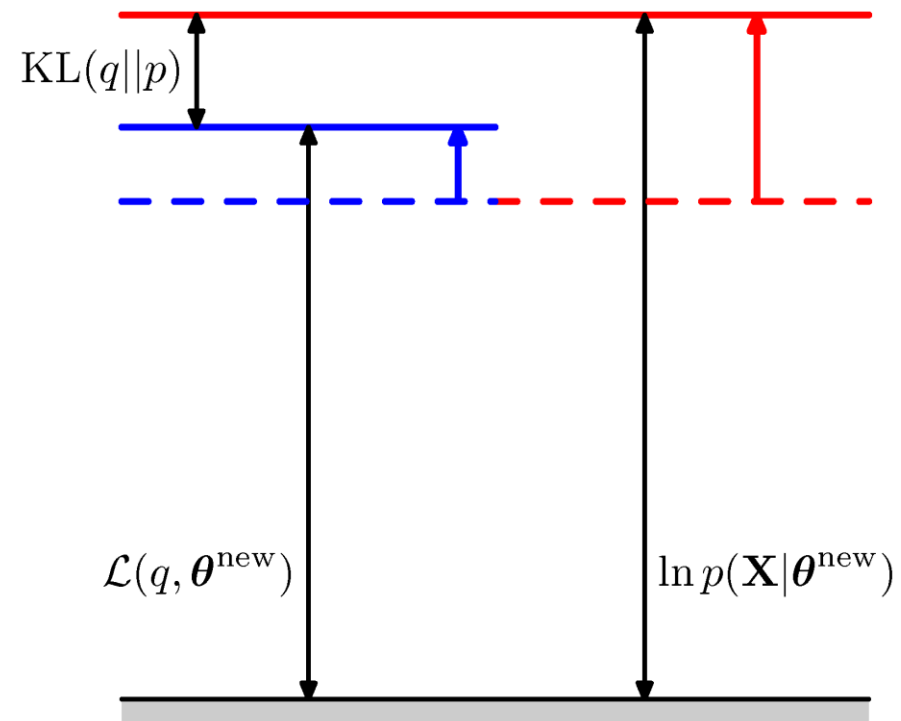
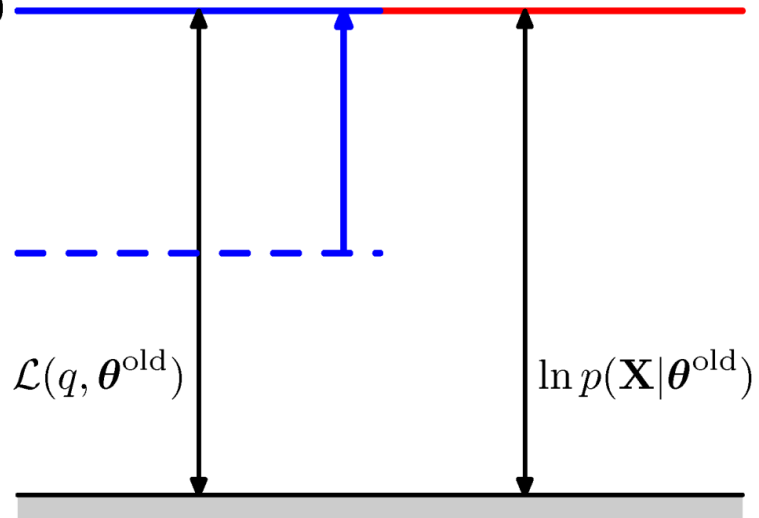
$$q(\mathbf{Z}) = p(\mathbf{Z}|\mathbf{X}, \theta^{old})$$

$$L(q, \theta) = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta) - \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{Z}|\mathbf{X}, \theta^{old}) = Q(\theta, \theta^{old}) + H(p)$$

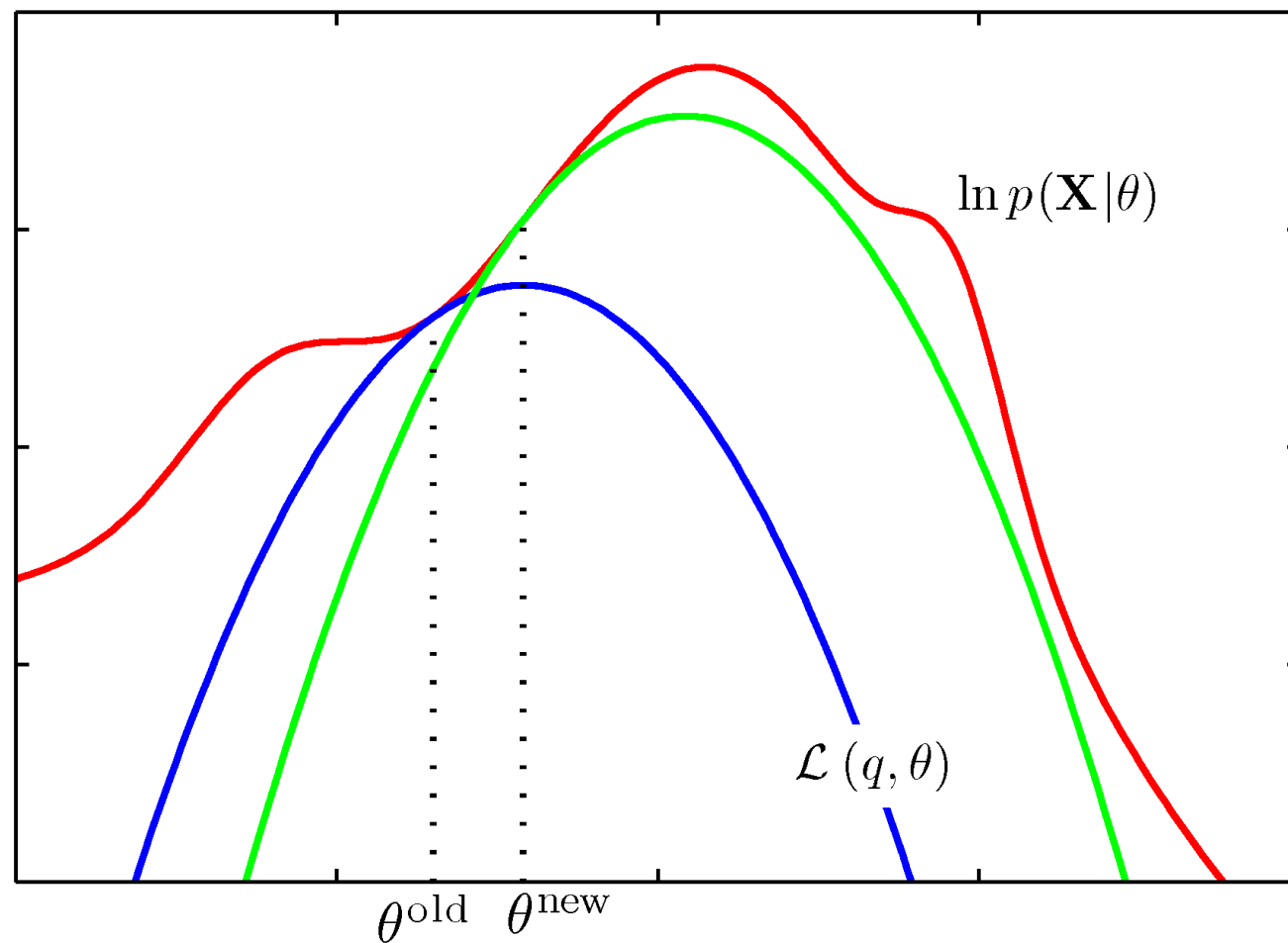
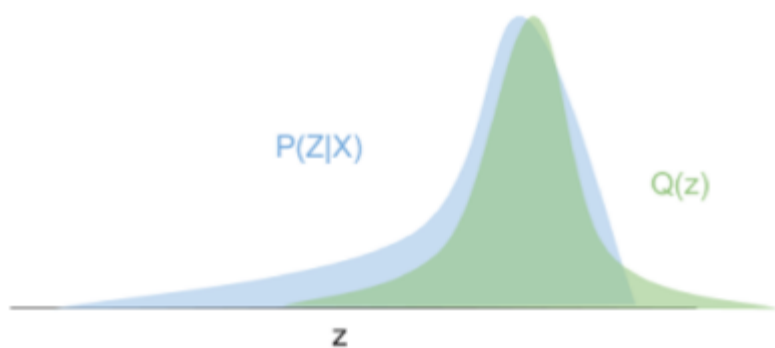
- we face Expectation of log expression
 - this is the reason that we use expectation of log rather than log-sum derived from the original GMM formulation
 - In other words, the reason we use Expectation of log is that this is the approach based on the variational inference
-

Variational Inference

$$\text{KL}(q||p) = 0$$



Variational Inference



Variational Inference

In summary: variational inference:

- To find the proper $q(Z|\theta')$ that can replace the intractable posterior $p(Z|X, \theta)$ of latent variable Z
- In general, we assume $q(Z|\theta')$ as the simple form such as uni-modal gaussian
- In the case of GMM, we can directly utilize posterior as q since the posterior is tractable

meaning:

- convert the probability distribution problem into optimization
 - provide the deterministic solution (cannot guarantee the global solution)
-
- is it possible to use more complex $q(Z|\theta')$ to catch the more complex (or intractable) distribution?
 - the distribution $p(Z|X, \theta)$ is highly related to data X ; is there any way to approximate q by using X ?
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Interim Summary

- Latent variables and GMM
- Variational Inference